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RESEARCH

A Versatile Mixed-Mode HPLC Method for Sensitive Quantification of Polyvinylpyrrolidone in Pharmaceutical, Cosmetic, and Environmental Samples with a Greenness Assessment

Samar M. Mahgoub[1] · Hassan A. Rudayni[2] · Abdelatty M. Radallada[3] · Heba A. Younes[4] · Ahmed Allam[2] · Rehab Mahmoud[3]

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Abstract

Purpose This study introduces an innovative mixed-mode high-performance liquid chromatography (HPLC) method employing a Hi-Plex H column that synergistically combines size exclusion and ion exchange mechanisms, enabling precise and reproducible quantification of PVP K30 in diverse matrices, including pharmaceutical, cosmetic, and environmental water.

Methods This study introduces an innovative mixed-mode high-performance liquid chromatography (HPLC) method employing a Hi-Plex H column using an isocratic 1:1 mix of 70% perchloric acid (pH ~1.5) and acetonitrile at 1.5 mL/min flow, 25°C, with detection at 210 nm and a 12-minute run time. A solid phase extraction pretreatment method was used to enhance their detection in wastewater. The method's environmental impact was rigorously audited using established greenness metrics (AGREE, AGREEprep, MoGAPI, BAGI, and RGB12), providing a transparent assessment of its sustainability profile.

Results The validated method demonstrated excellent linearity (4-500 µg/mL; r^2= 0.9997) with limits of detection and quantification of 1.2 µg/mL and 3.8 µg/mL, respectively. Implementation of an optimized SPE protocol, providing a 500-fold preconcentration, enabled exceptional environmental sensitivity (LOD: 2.4 ng/L in the original water sample). High accuracy was achieved with recovery rates of 98.35-99.45% in commercial products and 82.02-85.10% in water samples. A comprehensive evaluation using AGREE, AGREEprep, MoGAPI, BAGI, and RGB12 tools provided a multi-faceted perspective on the method's practicality and environmental footprint, with a high practical applicability score (BAGI = 70.0) and an overall whiteness score of 72.0.

Conclusion The developed method represents a significant advancement in polymer analysis, providing a sensitive and versatile solution for PVP K30 monitoring in both industrial quality control and environmental protection applications. The integrated greenness assessment provides a benchmark for the environmental impact of this analysis and identifies areas for future improvement.

✉ Rehab Mahmoud
rehabkhaled@science.bsu.edu.eg

Samar M. Mahgoub
miramar15@yahoo.com

Hassan A. Rudayni
HARUDAYNI@imamu.edu.sa

Abdelatty M. Radallada
abdelaty.mohamed@science.bsu.edu.eg

Heba A. Younes
h.ahmed@psas.bsu.edu.eg

Ahmed Allam
aallam@imamu.edu.sa

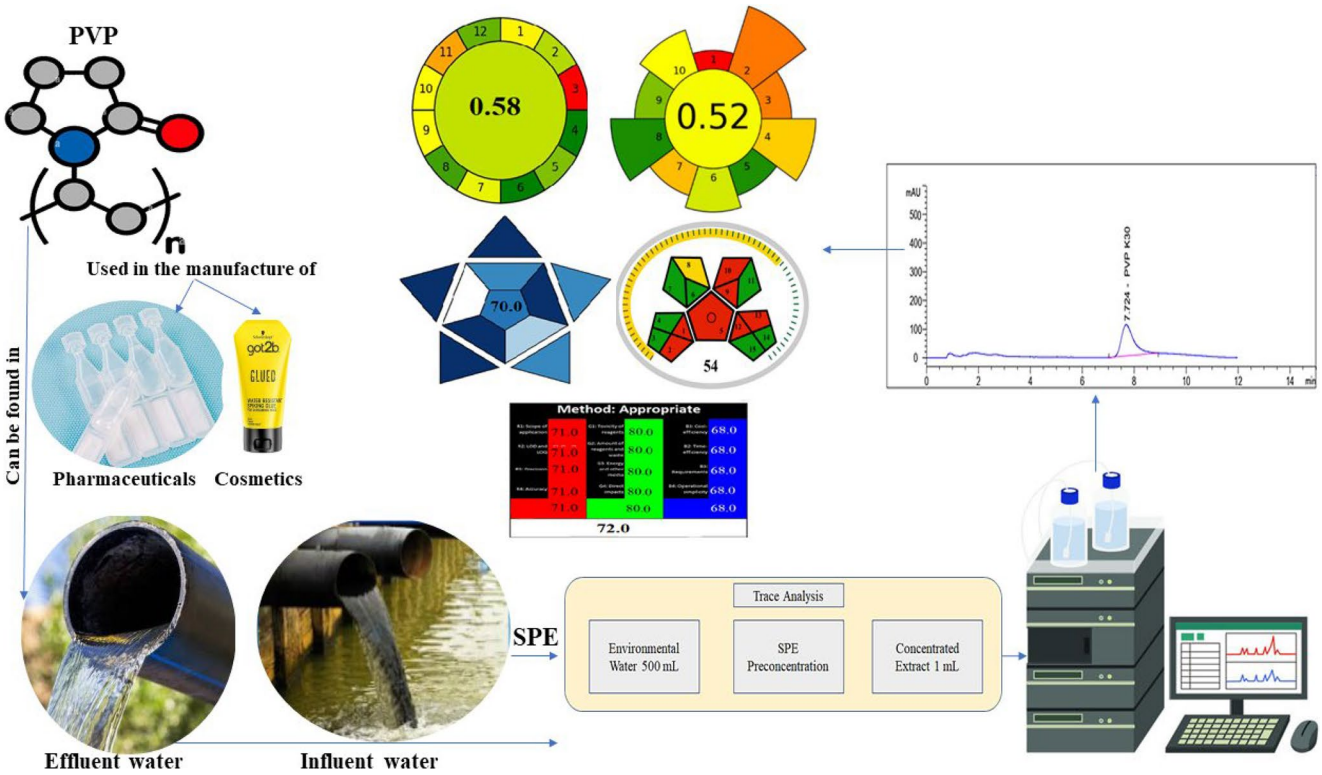
[1] Materials Science and Nanotechnology Department, Faculty of Postgraduate Studies for Advanced Sciences, Beni-Suef University, Beni-Suef, Egypt

[2] Department of Biology, College of Science, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh 11623, Saudi Arabia

[3] Department of Chemistry, Faculty of Science, Beni-Suef University, Beni-Suef 62528, Egypt

[4] Environmental Science and Industrial Development Department, Faculty of Postgraduate Studies for Advanced Sciences, Beni-Suef University, Beni-Suef, Egypt

Graphical Abstract



Keywords Polyvinylpyrrolidone K30 · Mixed-mode HPLC · Method validation · Pharmaceutical analysis · Environmental monitoring · Greenness assessment

Introduction

The ubiquitous presence of synthetic polymers in commercial products and their subsequent entry into aquatic ecosystems has emerged as a significant environmental and regulatory concern over the past decade [1]. Among these polymers, polyvinylpyrrolidone (PVP), particularly the K30 grade, has gained attention because of its widespread use in both consumer products and industrial applications (Fig. 1) [2, 3]. PVP K30 is extensively utilized as an excipient in pharmaceutical formulations, a stabilizing agent in cosmetics, a binding agent in industrial applications, and numerous consumer products [4]. The versatility of PVP K30 has led to its incorporation into a wide range of commercial products, including ophthalmic solutions, hair styling products, tablet formulations, and topical preparations [5]. Moreover, the increasing consumption of PVP-containing products has led to their detection in surface water, groundwater, and wastewater treatment plant effluents,, with concentrations ranging from ng/L to low µg/L [6–11]. These findings demonstrate the persistence of PVP due to its resistance to conventional wastewater treatment processes and raise concerns about its potential environmental impacts. Recent ecotoxicological assessments suggest that PVP can accumulate in aquatic organisms and may influence microbial community structure [10, 12]. Despite its classification as generally recognized as safe (GRAS) for human consumption, the ecological implications of persistent PVP exposure remain incompletely understood, necessitating improved analytical tools for environmental monitoring [13, 14].

Although PVP is safe for human use, its high-water solubility, resistance to biodegradation, and ability to bind contaminants raise concerns about persistence and transport in aquatic systems. Its incomplete removal in wastewater treatment highlights continuous environmental release, and the unknown long-term effects of chronic exposure underscore the need for sensitive monitoring tools, which the developed method provides for lifecycle assessment, current analytical methods for PVP quantification, particularly in environmental samples and complex product matrices, are often hampered by inadequate sensitivity for trace-level detection, insufficient selectivity in complex matrices, poor selectivity and significant interference effects [15, 16]. This creates a need for analytical methodologies to track PVP throughout its lifecycle from commercial formulations to the

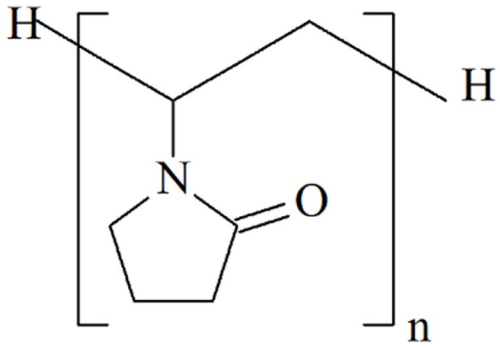


Fig. 1 Chemical structure of polyvinylpyrrolidone (PVP K30)

environment. Effective quality control of PVP-containing products and environmental monitoring requires sensitive, specific, and validated analytical methods. While several techniques have been employed for polymer detection, including spectroscopic methods [17–19], size exclusion [20], paper chromatography [21], and gas chromatography [22], these approaches often lack the sensitivity required for environmental samples or suffer from matrix interference effects in complex product formulations [23]. High-performance liquid chromatography (HPLC) offers advantages in terms of sensitivity, reproducibility, and applicability to complex matrices in both pharmaceutical and cosmetic products and environmental samples [24, 25]. Additionally, using pretreatment methods for samples with high matrix before HPLC analysis are essential to improve the accuracy, sensitivity, and reliability of results. solid phase extraction (SPE) being one of the most effective techniques because it is less time-consuming and uses less solvent compared to traditional methods like liquid-liquid extraction, making it more cost-effective and environmentally friendly [26]. Previous attempts to quantify PVP via HPLC have employed C18 columns with various mobile phase compositions, yet challenges persist regarding retention and quantification limits across diverse applications [27].

On the other hand, using HPLC causes environmental problems because most mobile phases are toxic organic solvents that produce a lot of hazardous waste. The assessment of greenness in chromatographic techniques is crucial. Different methods were applied for this assessment, such as twelve green analytical chemistry principles (GAC) [28]. These guiding principles safeguard both the environment and human safety by waste reduction, atom economy, safer chemicals, energy efficiency, renewable feedstock, and design for degradation [29]. To comprehensively evaluate the environmental impact and sustainability of our method, we applied multiple modern assessment tools. The AGREE quantifies compliance with the 12 principles of green analytical chemistry and AGREEprep specifically assesses the greenness of sample preparation [30]. The Modified Green Analytical Procedure Index (MoGAPI) extends GAPI by offering a more detailed lifecycle evaluation [31], and the Blue Applicability Grade Index (BAGI) complements these by rating practicality, cost, and efficiency [32]. Finally, the RGB 12 algorithm integrates analytical efficacy (red), ecological impact (green), and practical/economic feasibility (blue) into a single "whiteness" score that reflects the overall sustainability and balance of the method [33]. Over time, focus has increased on using green tools to assess the environmental evaluation of pharmaceutical analysis methods. Developing unified analytical methods optimized explicitly for PVP K30 detection across its product-to-environment lifecycle remains an ongoing need in analytical chemistry. To overcome these challenges, we developed a novel mixed-mode HPLC method utilizing a Hi-Plex H column. This stationary phase synergistically combines size exclusion and ion exchange mechanisms, providing a unique selectivity that is particularly suited for the analysis of challenging polymers like PVP K30. The novelty of this work is fourfold where It is the first reported use of a mixed-mode (size-exclusion/ion-exchange) HPLC mechanism for the analysis of PVP K30, overcoming the retention issues of reversed-phase methods. It presents a single, validated method applicable across three distinct matrices: pharmaceuticals, cosmetics, and environmental water, which has not been achieved before. It achieves exceptional sensitivity for environmental monitoring (LOD 2.4 ng/L) through an optimized SPE preconcentration protocol. It incorporates a comprehensive multi-metric greenness assessment to rigorously evaluate and validate the method's sustainability. This study aims to bridge a significant methodological gap by providing a sensitive, versatile, and more sustainable solution for PVP K30 monitoring.

Materials and Methods

Chemicals and Reagents

The polyvinylpyrrolidone K30 raw material (average MW 55,000 Da, pharmaceutical grade, ≥95% purity) was purchased from Sigma–Aldrich (Steinheim, Germany). HPLC-grade perchloric acid (70%, purity≥99.9%) and HPLC-grade acetonitrile (purity≥99.9%) were procured from Merck (Darmstadt, Germany). HPLC-grade water was produced using a Milli-Q purification system (Millipore, Bedford, MA, USA). Refresh® Classic eye drops containing 0.6% povidone (PVP K30) per single container (0.4 mL) were manufactured by AbbVie Inc. (North Chicago, IL, USA), and Got2B Glued Spiking Glue was manufactured by Henkel Corporation (Düsseldorf, Germany). All other chemicals and solvents used were of analytical grade (purity≥99.0%) or HPLC grade and were obtained from reputable commercial sources.

Sample Collection

Environmental water samples (n >20) were collected from over 20 distinct water bodies across Egypt between January and March 2024. Influent and effluent wastewater samples were collected from the Al-Gabal Al-Asfar treatment plant in Cairo (30°13′N, 31°25′E) [34]. All samples were collected in precleaned 1 L amber glass bottles via established environmental sampling protocols [35, 36], they were preserved at 4 °C, transported to the laboratory within 6 h, and filtered through 0.22 µm nylon membrane filters to remove particulate matter prior to analysis.

Instrumentation and Chromatographic Conditions

The chromatographic experiments were conducted via an Agilent 1200 Series HPLC system (Agilent Technologies, California, USA). This system was configured with a quaternary pump (model G1311A), an integrated degasser (G1322A), an autosampler (G1329A), a column compartment with temperature control (G1316A), and a variable-wavelength UV detector (G1314B). Data collection and analysis were performed with ChemStation software (version B.04.03, Agilent Technologies). For separation, a Hi-Plex H USP L17 column (100 mm × 7.7 mm, 8 µm particle size; Agilent Technologies) was utilized. The mobile phase was prepared by mixing two components: Component A was prepared by diluting 1 mL of 70% perchloric acid in 1 L of HPLC-grade water, resulting in a pH of approximately 1.5. Component B was pure HPLC-grade acetonitrile. Equal volumes of both components (1:1, v/v) were combined immediately before use and degassed by sonication for 10 min. Isocratic elution was applied throughout the analysis, maintaining a constant flow rate of 1.5 mL/min. This flow rate was selected as it provided an optimal balance between analysis time and chromatographic efficiency (theoretical plates > 2500), generating a system backpressure of approximately 120 bar, which is within the operational limits of the column and instrument. The sample injection volume was fixed at 100 µL. This relatively large injection volume was selected to enhance method sensitivity and achieve lower limits of detection, which was particularly important given the weak UV absorbance of PVP K30 at 210 nm. Method development experiments confirmed that this injection volume provided optimal sensitivity without compromising peak shape or chromatographic resolution. The column was kept at a stable temperature of 25 °C. Detection was performed at 210 nm, corresponding to the maximum absorbance for PVP K30. Each run lasted for 12 min, the PVP K30 peak typically observed at approximately 7.7 min.

Preparation of Standard Solutions

To prepare the stock solution, 100 mg of PVP K30 was weighed precisely and dissolved in 100 mL of distilled water, resulting in a concentration of 1000 µg/mL. The solution was subjected to sonication for 10 min to ensure that it was fully dissolved and then stored at 4 °C until use. Each day, fresh working standard solutions were made by diluting the stock solution with distilled water to achieve final concentrations of 4, 20, 40, 80, 200, 400, and 500 µg/mL.

Construction of the Calibration Curve

Serial dilutions ranging from 4 to 500 µg/mL were prepared from a 1000 µg/mL stock solution of PVP K30 using distilled water as the diluent. For each matrix type, both solvent-based and matrix-matched calibration curves were constructed to evaluate potential matrix effects and ensure accurate quantification across different sample types. Solvent-based calibration: Standard solutions were prepared in distilled water as the reference calibration approach. Matrix-matched calibration: To account for potential matrix effects, calibration standards were prepared in representative blank matrices for each application: Pharmaceutical matrix: Blank eye drop formulation (confirmed PVP-free) was processed according to Sect. 2.7.1 and spiked with known concentrations of PVP K30 (4, 20, 40, 80, 200, 400, and 500 µg/mL). Cosmetic matrix: Blank hair gel formulation (confirmed PVP-free) was processed similarly and spiked at equivalent concentrations. Environmental matrix: Blank environmental water samples (confirmed to contain < LOD of PVP K30) were processed through the complete SPE procedure described in Sect. 2.7.2 and subsequently spiked at corresponding concentration levels. Each concentration was analyzed in triplicate under the optimized chromatographic conditions described in Sect. 2.3. The peak areas were recorded at 210 nm and plotted against the corresponding concentrations to construct both solvent-based and matrix-matched calibration curves. Linear regression analysis was performed to determine the linearity parameters for each approach. Matrix effect evaluation: Matrix effects were quantitatively assessed by comparing the slopes of matrix-matched calibration curves with those of solvent-based calibration curves using Eq. (1):

$$Matrix\ Effect\ (\%) = \frac{Slope_{Matrix-matched} - Slope_{solvent-based}}{Slope_{solvent-based}} \quad (1)$$

Matrix effects within the range of −20% to +20% were considered acceptable, in accordance with quantitative bioanalytical method guidelines.

Method Validation

The newly established mixed-mode HPLC method was validated in accordance with the International Conference on Harmonization (ICH) Q2(R1) guidelines [37], which address the following performance characteristics:

Linearity and Range

The linearity of the method was evaluated using both solvent-based and matrix-matched calibration approaches by preparing seven standard solutions of PVP K30 across a concentration range of 4–500 µg/mL (4, 20, 40, 80, 200, 400, and 500 µg/mL), with each concentration analyzed in triplicate under the optimized chromatographic conditions. Solvent-based calibration: Standards were prepared in distilled water. Matrix-matched calibration: To account for potential matrix effects, standards were prepared in representative blank matrices for pharmaceuticals (diluted blank eye drops), cosmetics (diluted blank hair gel), and environmental samples (blank water processed through SPE). The calibration range was deliberately set with a lower limit of 4 µg/mL, positioned just above the pre-determined Limit of Quantification (LOQ = 3.8 µg/mL), to ensure accurate and precise quantification at trace levels. This upper bound of 500 µg/mL was selected to fully encompass the expected concentrations of prepared samples following simple dilution of pharmaceutical and cosmetic products, thereby minimizing unnecessary dilution steps and potential error introduction. Furthermore, the range was designed to comprehensively bracket the anticipated concentrations in all sample matrices after preparation: environmental samples, following solid-phase extraction (SPE) with a 500-fold preconcentration, yield concentrations within this quantifiable range, while diluted product samples consistently fall within the mid-to-upper calibration span. The selected range thereby adheres to ICH Q2(R1) guidelines by adequately covering expected sample concentrations while demonstrating acceptable linearity, accuracy, and precision throughout. Calibration curves were constructed by plotting the mean peak area against PVP K30 concentration for both approaches, and linear regression analysis was performed to determine the slope, y-intercept, and correlation coefficient (r^2). Matrix effects were quantitatively assessed by comparing the slopes of matrix-matched versus solvent-based calibrations (Eq. 1) to ensure method accuracy across all target matrices. The results confirmed method performance across the validated range for all matrices, with minimal matrix effects observed (see Sect. 3.2.1, Table 1A).

Furthermore, to ensure the absence of systematic bias and carry-over, additional parameters were evaluated following established chromatographic validation guidelines. The y-intercept of the solvent-based calibration curve was expressed as a percentage of the response of the target analyte at the LOQ level, yielding a value of 4.3%, which is well below the recommended threshold of 20–25% [38], confirming that the method is free from significant bias at low concentrations. Carry-over was assessed by injecting a blank solvent sample immediately after the highest calibration standard (500 µg/mL), a methodology aligned with best practices for identifying injection-to-injection contamination [39]. No peak area greater than 20% of the LOQ peak area was detected at the retention time of PVP K30, demonstrating that carry-over is negligible and does not impact the quantitation of subsequent samples [40]. This rigorous assessment ensures the method's reliability for routine analysis across pharmaceutical, cosmetic, and environmental matrices.

Table 1 System suitability parameters for PVP K30 determination ($n = 6$)

Parameter	Measured Value	Acceptance Criteria
Retention time (min)	7.70 ± 0.05	RSD ≤ 1.0%
Theoretical plates	2526 ± 87	> 2000
Tailing factor	1.12 ± 0.03	≤ 2.0
Capacity factor (k')	3.48	> 2.0 (indicates good retention)
Selectivity factor (α)	N/A	N/A (only one analyte peak)

Precision

The precision of the method was assessed at three levels. For repeatability (intraday precision), six replicate measurements were conducted for each of three PVP K30 concentrations (low: 10 µg/mL, medium: 80 µg/mL, and high: 400 µg/mL) within a single day. To evaluate intermediate precision (interday precision), the same concentration levels were analyzed over three consecutive days. The results are expressed as the relative standard deviation (RSD) of the measured values.

Accuracy

The accuracy of the method was determined using matrix-matched calibration curves and recovery studies conducted at three different concentrations for each matrix type. This approach ensures that potential matrix effects are properly accounted for during quantification, providing more reliable and accurate results than solvent-based calibration alone. For pharmaceutical and cosmetic products, known amounts of PVP K30 were spiked into representative blank matrices at 80%, 100%, and 120% of the target assay concentration, in accordance with ICH Q2(R1) guidelines [37].

The quantification was performed using matrix-matched calibration curves prepared in the corresponding blank matrices processed identically to real samples. For environmental water samples, blank environmental water samples (confirmed to contain < LOD of PVP K30) were spiked at concentrations of 20, 40, and 80 µg/L (pre-SPE) and processed through the complete SPE procedure described in Sect. 2.7.2. Quantification was performed using matrix-matched calibration curves prepared from blank water samples processed identically through the SPE protocol. Each concentration level was analyzed in triplicate using the corresponding matrix-matched calibration curve to ensure statistical reliability. The recovery percentage was calculated via the following Eq. (7):

$$\frac{Recovery\ (\%) = \text{Measured concentration after spiking} - \text{Concentration of unspiked sample}}{\text{Amount spiked}} \times 100 \quad (2)$$

The acceptance criteria were set in accordance with ICH Q2(R1) guidelines for pharmaceutical products [34] (recovery range of 98–102% and RSD ≤ 2% for the 80%, 100%, and 120% levels) and US EPA guidelines for environmental analysis [41] (recovery range of 80–120% and RSD ≤ 5% for complex matrices), to ensure method accuracy and precision across different concentration levels and matrices.

Limits of Detection and Quantification

The limits of detection (LODs) and quantifications (LOQs) were determined via the standard deviation of the response (σ) and the slope of the calibration curve (S). These values were calculated according to the following formulas:

$$LOD = 3.3 \times (\sigma/S) \quad (3)$$

$$LOQ = 10 \times (\sigma/S) \quad (4)$$

For pharmaceutical and cosmetic product samples, the LOD and LOQ were determined directly from standard solutions and spiked product matrices and are expressed in terms of µg/mL in the analyzed solution. For environmental water samples, the method includes a solid-phase extraction (SPE) preconcentration step, in which a large sample volume (typically 500 mL) is concentrated and reconstituted into a smaller final volume (typically 1 mL). To express the LOD and LOQ relative to the original environmental sample, the values obtained from the extract (µg/mL) were converted via the following equations:

$$LOD_{env}\left(\frac{\mu g}{L}\right) = \frac{LOD_{extract}\left(\frac{\mu g}{mL}\right)}{Pre-concentration\ factor} \times 1000 \quad (5)$$

$$LOQ_{env}\left(\frac{\mu g}{L}\right) = \frac{LOQ_{extract}\left(\frac{\mu g}{mL}\right)}{Pre-concentration\ factor} \times 1000 \quad (6)$$

where the preconcentration factor is defined as the ratio of the initial sample volume (mL) to the final extract volume (mL). This approach ensures that the method's sensitivity is appropriately translated to the original environmental sample concentration. Notably, owing to the preconcentration protocol, trace-level environmental analytes present at low µg/L concentrations in large water samples can be brought within the quantifiable range of the HPLC method (expressed as µg/mL). This preconcentration approach is a fundamental and validated technique in environmental analytics designed to bridge the gap between trace-level concentrations in the original sample and the working range of chromatographic instruments. The calibration range (4–500 µg/mL) applies to the concentration in the final extract, while the reported environmental LOD and LOQ (ng/L) are calculated for the original water sample prior to preconcentration.

Specificity

The specificity of the method was assessed by comparing chromatograms of blank samples (distilled water, mobile phase and blank environmental samples), PVP K30 standard solutions, commercial product samples, and spiked water samples to evaluate potential interference from matrix components across different applications.

Robustness

To assess the robustness of the method, key parameters were intentionally altered, including adjusting the flow rate by ± 0.1 mL/min, changing the detection wavelength by ± 2 nm, and modifying the acetonitrile content in the mobile phase by ± 1%. The influence of these variations on both the retention time and peak area was systematically investigated.

System Suitability

To confirm the adequacy of the chromatographic system, key suitability parameters were evaluated in accordance with chromatographic guidelines. This assessment was performed by injecting a 200 µg/mL PVP K30 standard solution six times consecutively. The parameters measured included retention time (t_R), number of theoretical plates (N), tailing factor (T_f), capacity factor (k'), and selectivity factor (α). The void time (t_0), determined by injecting an uracil solution, was 1.72 min. The resulting data are presented in Table 1, and a typical chromatogram is shown in Fig. 2.

Ruggedness

The ruggedness of the method was evaluated to determine its reproducibility under varied conditions, a critical factor for its transferability and use in quality control laboratories. The study was designed in accordance with ICH Q2(R1) guidelines and involved two different analysts using two different HPLC systems (Agilent 1200 and Agilent 1260 Infinity II) on two different days. Each analyst prepared a standard solution of PVP K30 at 200 µg/mL independently and injected it six times ($n=6$) under the established chromatographic conditions. The retention time, peak area, and theoretical plates were recorded and compared. The results were evaluated using the relative standard deviation (RSD) between the sets of data generated by the two analysts and systems.

Sample Analysis

Analysis of Pharmaceutical and Cosmetic Products

For the analysis of Refresh® Classic eye drops, the contents of a single-dose container (0.4 mL, containing 0.6% PVP K30) were transferred into a 10 mL volumetric flask and diluted to the mark with distilled water, resulting in a theoretical PVP K30 concentration of 0.24 mg/mL. In the case of Got2B-Glued Spiking Glue, approximately 1 g of the product was accurately weighed and placed in a 100 mL volumetric flask. The sample was dissolved in approximately 50 mL of distilled water with the aid of 15 min of sonication and then diluted to volume with distilled water to ensure that the final concentration fell within the calibration range. The sample preparation for pharmaceutical and cosmetic products was designed based on the principles of minimum manipulation and maximum compatibility with the chromatographic system. Simple dilution with water was selected as it effectively dissolves PVP K30, reduces matrix complexity (e.g., salt content, viscosity), and brings the analyte concentration within the validated linear range (4–500 µg/mL). This approach is aligned with ICH Q2(R1) guidelines, which recommend simplicity and robustness in sample preparation for pharmaceutical analysis [42]. The selected dilution factors were calculated based on the labeled or expected PVP content to target a mid-range concentration (200 µg/mL), optimizing accuracy and precision while minimizing potential matrix effects and the need for extensive extraction procedures commonly reported for complex formulations [27]. Both prepared solutions were filtered through 0.22 µm nylon membrane filters prior to HPLC injection. To evaluate method accuracy in these matrices, recovery experiments were conducted by spiking the diluted samples with a known amount of PVP K30 (0.1 mg/mL) and analyzing the mixtures. Each determination was performed in triplicate to provide statistically valid results.

Analysis of Environmental Water Samples

For the quantification of PVP K30 in environmental water, a solid-phase extraction (SPE) protocol was employed using Oasis HLB cartridges (6 mL, 200 mg, Waters Corp., USA). This sorbent was selected due to its hydrophilic-lipophilic balanced chemistry, which provides excellent retention for polar, water-soluble compounds like PVP K30. The N-vinylpyrrolidone component of the HLB copolymer offers particular affinity for PVP through hydrogen bonding and polar interactions, ensuring high extraction efficiency. The solid-phase extraction (SPE) protocol using Oasis HLB cartridges was optimized and selected based on its well-documented high retention capacity for a wide range

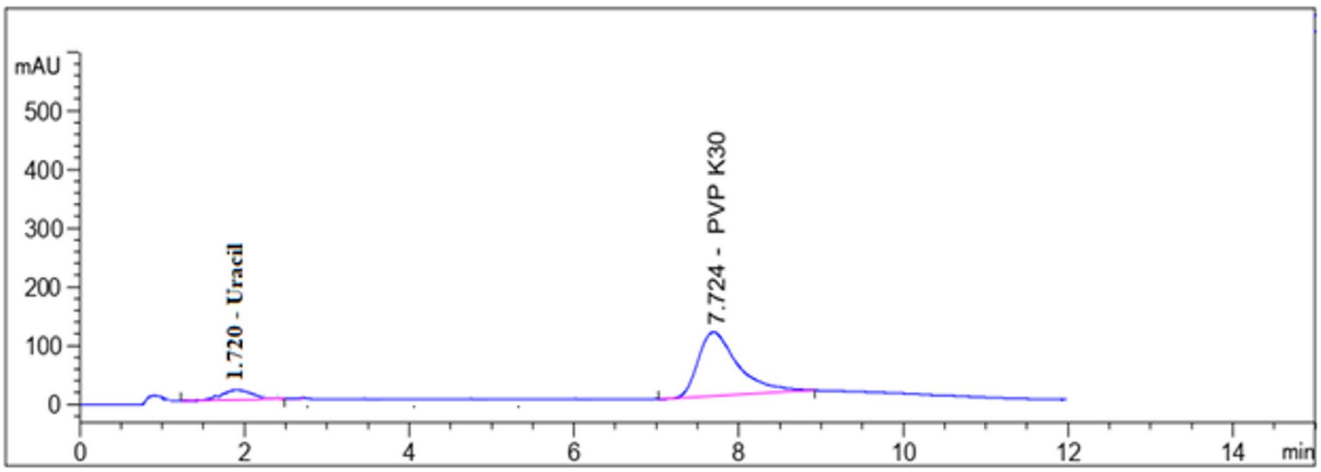


Fig. 2 Representative HPLC chromatogram of a PVP K30 standard solution (200 µg/mL) recorded at a wavelength of 210 nm. The unretained solvent front (t_0) is observed at 1.72 min. The PVP K30 peak elutes at approximately 7.70 min

of polar and non-polar compounds, making it a standard choice in environmental analytical chemistry for the extraction of diverse organic micropollutants from water [26, 41]. The hydrophilic-lipophilic balanced (HLB) sorbent is particularly effective for retaining water-soluble polymers like PVP K30 through a combination of hydrophobic interactions and hydrogen bonding with the pyrrolidone rings. The sample volume (500 mL), elution solvent (methanol), and preconcentration factor (500x) were optimized to achieve the required sensitivity for trace-level environmental analysis (target ng/L levels). This approach is consistent with US EPA methodologies for the analysis of organic contaminants in water, which often employ large-volume extraction and preconcentration to meet low detection limits [41]. The SPE cartridges were preconditioned by sequentially passing 5 mL of methanol and 10 mL of distilled water through them at a flow rate of approximately 1 mL/min. Filtered water samples (500 mL each) were then loaded onto the cartridges at a rate of 5 mL/min using a vacuum manifold, allowing for the retention of PVP K30. After sample loading, the cartridges were rinsed with 5 mL of distilled water to remove weakly bound substances.

In alignment with green chemistry principles and to address sustainability, ethanol was rigorously evaluated as a potential alternative to methanol for the elution stage. However, under the optimized conditions of this method, ethanol demonstrated a significantly lower elution strength, resulting in poor and inconsistent recovery of PVP K30 (approximately 72%). As achieving high and consistent recovery is paramount for the accurate quantification of trace-level analytes, methanol was retained as the elution solvent to ensure method robustness, reproducibility, and to meet our stringent validation criteria. Future work will focus on optimizing a protocol with a greener solvent without compromising this critical performance parameter.

PVP K30 was eluted with 5 mL of methanol. The methanolic eluate was collected in clean glass tubes, evaporated to near dryness at 40 °C under a gentle stream of nitrogen, and reconstituted in 1 mL of distilled water. This process resulted in a preconcentration factor of 500, which was calculated via the following equation:

$$\text{Pre}-\text{concentration factor (PF)} = \frac{V_{Sample}}{V_{Final\ extract}} = \frac{500\ mL}{1\ mL} = 500\times \quad (7)$$

The final concentration was filtered through a 0.22 µm nylon syringe filter and transferred to HPLC vials for subsequent analysis. To correlate the concentration measured in the final extract (in µg/mL) to the original environmental concentration (in µg/L), the following conversion equation was applied:

$$C_{original}\left(\frac{\mu g}{L}\right) = \frac{C_{extract}\left(\frac{\mu g}{mL}\right)}{PF} \times 1000 = \frac{C_{extract}}{500} \times 1000 = C_{extract} \times 2 \quad (8)$$

where $C_{extract\ is\ the}$ concentration of PVP K30 in the reconstituted SPE extract (µg/mL), as determined by HPLC; 1000 = the unit conversion factor from mL to L; and PF = 500 (preconcentration factor).

This equation enables back-calculation of the true PVP K30 concentration in the original environmental water sample before enrichment. All reported environmental concentrations were corrected via this formula to ensure accurate trace-level quantification (Scheme 1).

Whiteness of the Method

The sustainability and environmental friendliness of the analytical methodology were thoroughly evaluated. Several assessment tools, including AGREE, AGREEprep, MoGAPI, BAGI, and RGB12, were employed to systematically examine the potential ecological impact of our analytical approach and its compatibility with green chemistry principles [43, 44].

Results and Discussion

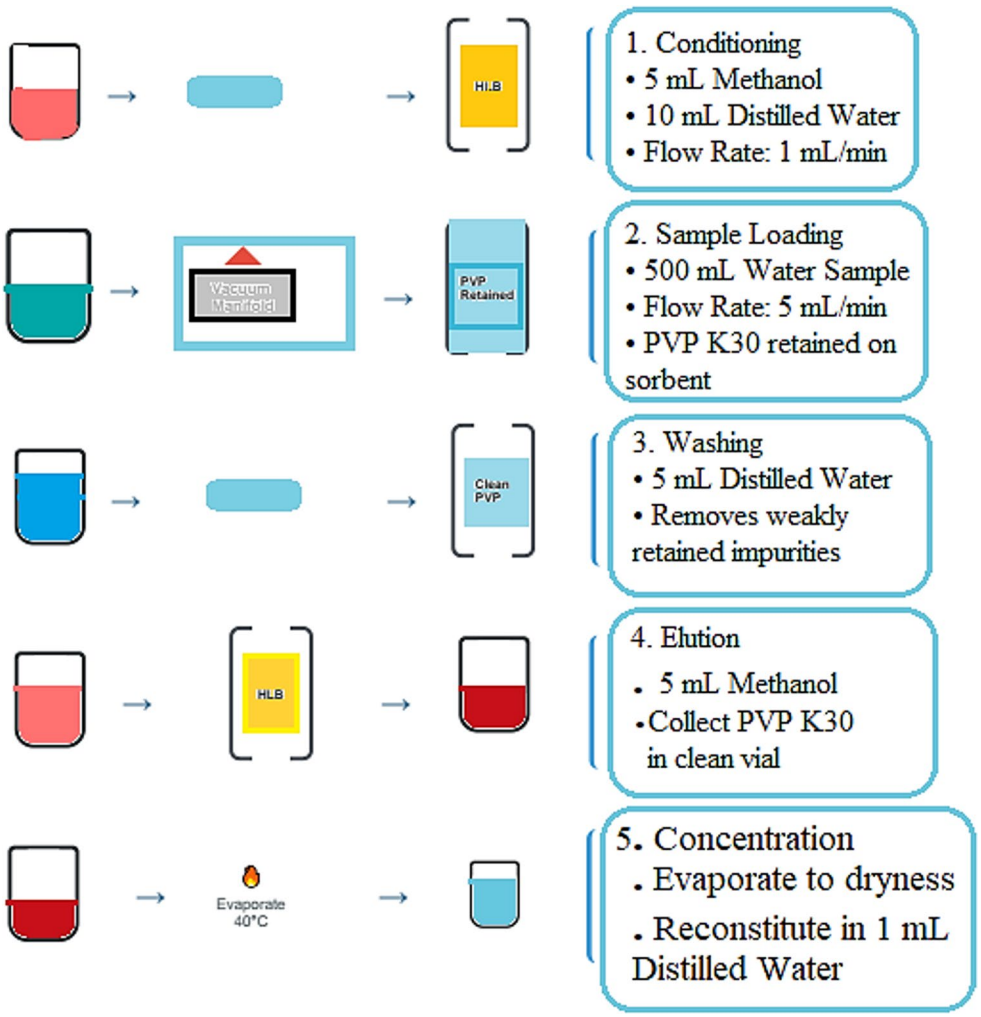
Method Development and Optimization

The development of an effective mixed-mode HPLC method for PVP K30 quantification presented several challenges due to the hydrophilic nature of the polymer and its limited UV absorbance characteristics. Initial method development focused on column selection, mobile phase composition, and detection parameters to achieve optimal sensitivity and selectivity. Chromatograms of trials and method validation were provided in supplementary Figure S1 A-F.

Selection of the Stationary Phase

The development of an effective HPLC method for PVP K30 quantification presented significant challenges due to the polymer's high hydrophilicity, lack of a chromophore, and broad molecular weight distribution. Initial method development focused on column selection to achieve optimal retention and peak shape.

Several conventional stationary phases were evaluated during preliminary experiments, including C18 (Zorbax Eclipse XDB-C18, 150 × 4.6 mm, 5 µm) and C8 (Agilent Eclipse XDB-C8, 150 × 4.6 mm, 5 µm) reverse-phase columns. As anticipated for a highly hydrophilic polymer, PVP K30 demonstrated negligible retention on these reverse-phase columns, eluting with or near the solvent front under a variety of mobile phase conditions. Initial screening of conventional reversed-phase columns, including C18 and C8 phases, proved ineffective for PVP K30 analysis. The performance



Scheme 1 Schematic representation of the solid-phase extraction (SPE) protocol used for preconcentration of PVP K30 from environmental water samples. The workflow includes cartridge conditioning, sample loading, washing, elution, solvent evaporation, and reconstitution, achieving a 500× preconcentration factor for trace-level quantification

was quantitatively assessed using a 200 µg/mL standard solution. As summarized in Table S1, both conventional columns demonstrated insufficient retention, with the analyte eluting near the solvent front (t_R >~ 2.2 min). More critically, they exhibited significant peak tailing (Tf >2.3) and low efficiency, which is a recognized limitation of reversed-phase chemistry for hydrophilic polymers, often due to undesirable secondary interactions with residual silanols [45, 46]. In stark contrast, the mixed-mode Hi-Plex H column provided optimal retention (t_R = 7.70 min), a symmetric peak shape (T_f = 1.12), and high efficiency (N >2500), resolving the critical limitations observed with other stationary phases. It is important to note that for a polydisperse polymer like PVP K30, the peak shape represents the molecular weight distribution. The symmetrical, Gaussian-shaped peak achieved with the Hi-Plex H column (Fig. 2) is indicative of a well-behaved chromatographic separation that effectively manages the polymer's polydispersity without the severe tailing caused by non-specific interactions on traditional reversed-phase columns. This symmetric peak profile is a significant advantage for accurate integration and quantification. This confirmed the fundamental incompatibility of standard reverse-phase mechanisms with PVP analysis.

The Hi-Plex H USP L17 column was ultimately selected for this method because its unique stationary phase chemistry directly addresses the analytical challenges posed by PVP. This column is packed with a sulfonated styrene-divinylbenzene copolymer, which provides a mixed-mode retention mechanism combining size exclusion and ion-exchange principles. This dual mechanism is particularly suited for PVP K30: the size exclusion mode helps manage the polymer's polydispersity, while the ion-exchange functionality (via the sulfonic acid groups) provides strong electrostatic interactions with the polar carbonyl groups of the pyrrolidone rings, especially under the acidic mobile phase conditions (pH ≈ 1.5) employed. This combination resulted in superior retention (tR = 7.70 min), a symmetric peak shape (Tf = 1.12), and high efficiency (N >2500), resolving the critical limitations observed with other stationary phases. This combination resulted in superior retention and a sharp, symmetric peak shape compared to all other columns tested, making it the clear choice for this application [47, 48].

The results confirm that conventional reversed-phase columns were unsuitable, as indicated by high tailing factors (Tf > 2.3), poor peak symmetry, and limited reproducibility. In contrast, the Hi-Plex H column exhibited superior performance (Tf = 1.12), offering adequate retention, high efficiency, and symmetric peaks. Its mixed-mode mechanism that is size exclusion to accommodate the polymer's polydispersity and ion-exchange to enhance interactions with pyrrolidone carbonyl groups under acidic conditions effectively overcomes the silanol interaction issues observed with standard reversed-phase columns, making it the optimal choice for PVP K30 analysis.

Mobile Phase Optimization

The mobile phase composition was systematically optimized to increase the sensitivity. The choice of acid was critical for promoting ionic interaction with the sulfonated stationary phase. Perchloric acid was selected over common alternatives like phosphoric acid or phosphate buffers due to its strong ion-pairing properties [49] and compatibility with UV detection at low wavelengths (210 nm). Its low UV cut-off and ability to provide a highly acidic environment (pH ~ 1.5) ensured efficient protonation of the PVP pyrrolidone carbonyl groups, maximizing retention and improving peak shape without introducing significant background absorbance [50]. Various concentrations of perchloric acid (0.5–2.0 mL in 1000 mL of water) were tested. The optimal composition was determined to be 1 mL perchloric acid in 1000 mL water, which provided the best compromise between retention time and peak symmetry. The acidic conditions enhanced the protonation of the carbonyl groups of PVP, improving the ionic interactions with the sulfonated stationary phase and resulting in sharper peaks. In alignment with green chemistry principles, the potential for replacing acetonitrile with a greener solvent was evaluated. In strict alignment with green chemistry principles, methanol was rigorously evaluated as the primary organic solvent due to its more favorable environmental and safety profile. However, its use resulted in several critical drawbacks under the mixed-mode conditions of the Hi-Plex H column. First, the higher viscosity of methanol-containing mobile phases resulted in a significantly elevated system backpressure (~ 180 bar compared to ~ 120 bar with acetonitrile), approaching the operational limits of the system and potentially compromising long-term column stability. Second, and most critically, methanol demonstrated a weaker elution strength for PVP K30 on this specific stationary phase, resulting in broader peaks and an approximately 30% reduction in peak area (and thus sensitivity) compared to acetonitrile. This loss in sensitivity would directly jeopardize the method's ability to detect trace levels of PVP in environmental samples, which is a core objective of this work. Ethanol was also investigated as a potential substitute. However, its use resulted in a significant increase in system backpressure (exceeding 200 bar) due to its higher viscosity, potentially compromising column longevity and requiring instrument modifications. More critically, chromatographic performance was adversely affected; ethanol led to broader peaks and a notable reduction in sensitivity (~ 30% decrease in peak area) for PVP K30, likely due to its weaker elution strength under the mixed-mode conditions. This loss in sensitivity would directly impact the method's ability to detect trace levels in environmental samples. Therefore, while acetonitrile is classified as less green, its use was deemed essential for achieving the required analytical performance (sensitivity, efficiency, and robustness) for this specific application. Therefore, while acetonitrile is classified as less green, its use was deemed essential to achieve the required analytical performance (sensitivity, efficiency, and robustness) for this specific application. To mitigate its environmental impact, the method employs an isocratic elution mode, which minimizes total solvent consumption and waste generation compared to gradient methods. Furthermore, all waste solvents are collected for proper disposal according to institutional environmental health and safety protocols. To mitigate its environmental impact, the method employs an isocratic elution mode, minimizing solvent consumption and waste generation compared to gradient methods.

Detection Wavelength Optimization

The UV absorbance of PVP K30 was evaluated across a range of wavelengths (200–300 nm). The maximum sensitivity was achieved at 210 nm, which is consistent with the $n \rightarrow \pi^*$ transitions of the carbonyl groups in the pyrrolidone rings. This wavelength provided adequate sensitivity while minimizing interference from common matrix components in water samples.

Flow Rate and Injection Volume

The effect of the flow rate (0.8–2.0 mL/min) on the separation efficiency was investigated. While a flow rate of 1.6 mL/min provided a marginally improved tailing factor (1.10), a flow rate of 1.5 mL/min was ultimately selected as the optimal flow rate. This decision was based on a holistic evaluation of chromatographic performance: the 1.5 mL/min rate provided an optimal balance between analysis time, peak shape, and system backpressure (~ 120 bar), whereas the higher flow rate significantly increased backpressure (~ 150 bar) while offering only a minimal reduction in analysis time (~ 0.9 min). The chosen flow rate provides

good peak shape and a reasonable analysis time (retention time of approximately 7.7 min for PVP K30) while ensuring long-term column stability. An injection volume of 100 µL was chosen to enhance the sensitivity without compromising peak symmetry or column efficiency.

Method Validation Results

System Suitability

The system suitability parameters were evaluated to ensure the adequate performance of the chromatographic system, and matrix effects were quantitatively assessed across all target matrices. The results are summarized in Table 1 and S2.

All system suitability parameters met the acceptance criteria, confirming the suitability of the chromatographic system for PVP K30 analysis. The low RSD value for retention time (0.65%) indicates excellent repeatability, while the high theoretical plate count (2526) demonstrates good column efficiency. The tailing factor of 1.12 indicates a symmetric peak shape, and the capacity factor of 3.48 confirms adequate retention away from the solvent front. The matrix effect evaluation demonstrates that matrix suppression is minimal across all three matrix types, with effects ranging from −1.24% to −3.24%, well within the acceptable range of ±20%. The slight signal suppression observed is consistent with the expected minor matrix effects in complex samples and does not significantly impact quantification accuracy. The high r^2 values (>0.999) for all matrix-matched calibrations confirm excellent linear response despite the presence of matrix components. These results validate the choice of matrix-matched calibration for accurate quantification across diverse sample types.

Linearity and Range

Excellent linearity was observed across the concentration range of 4–500 µg/mL for both solvent-based and matrix-matched calibration approaches. The results are summarized in Table S3 and illustrated in Fig. 3A and B.

The high r^2 values (greater than 0.999) confirm excellent linear response of the method throughout the evaluated range for all calibration approaches. The matrix-matched calibrations demonstrate that the presence of matrix components causes minimal signal suppression (1.24–3.24%), which is well within acceptable limits for bioanalytical methods. The slightly higher LOD and LOQ values for matrix-matched calibrations reflect the minor matrix effects but remain adequate for all intended applications. For environmental applications where PVP K30 concentrations typically range from ng/L to low µg/L, the solid-phase extraction preconcentration step (500×) effectively brings analyte concentrations within the quantifiable range. The calculated environmental LOD (1.5 µg/mL) and LOQ (4.6 µg/mL) in the matrix-matched extract translate to environmental detection limits of 3.0 ng/L (LOD) and 9.2 ng/L (LOQ) in the original water samples, maintaining exceptional sensitivity for trace-level environmental monitoring while accounting for matrix effects.

Limits of Detection and Quantification

Using the standard deviation of the response and the slope of the calibration curve, the method's limit of detection (LOD) and limit of quantification (LOQ) were established as 1.2 µg/mL and 3.8 µg/mL, respectively. The calculated LOD (1.2 µg/mL) and LOQ (3.8 µg/mL) represent the instrument detection limits for direct injection. For environmental applications, where PVP K30 concentrations typically range from ng/L to low µg/L, a solid-phase extraction preconcentration step (500×) was implemented to bring analyte concentrations within the quantifiable range. This preconcentration approach effectively lowers the method detection limit to 2.4 ng/L (LOD) and 7.6 ng/L (LOQ) in the original environmental samples, making the method suitable for trace-level environmental monitoring. The preconcentration factor was validated through recovery studies using spiked ultrapure water samples [51–53].

Precision

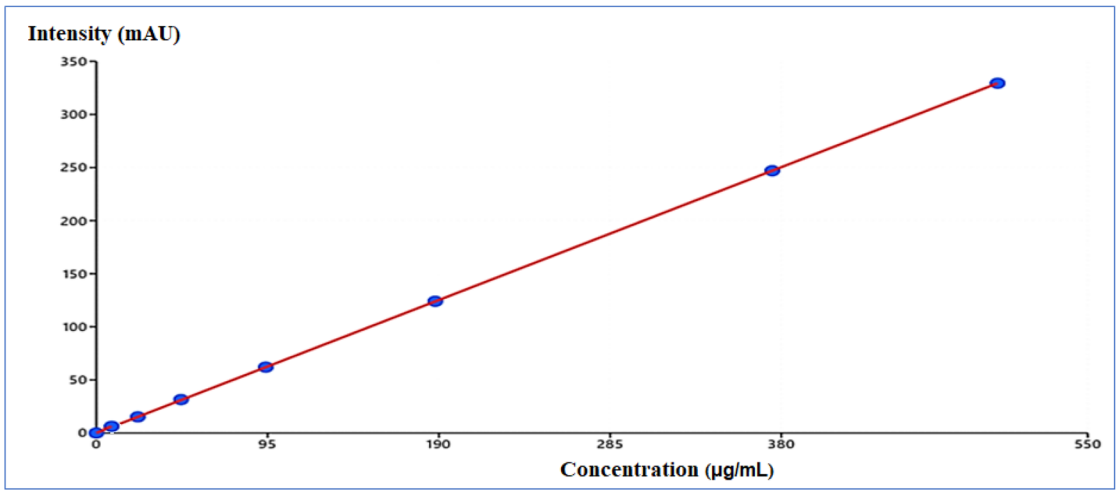
The method demonstrated excellent precision at all tested concentrations. The results for repeatability (intraday precision) and intermediate precision (interday precision) are summarized in Table 2.

The RSD values for both intraday and interday precision were well below 3%, indicating excellent precision across the tested concentration range [54, 55].

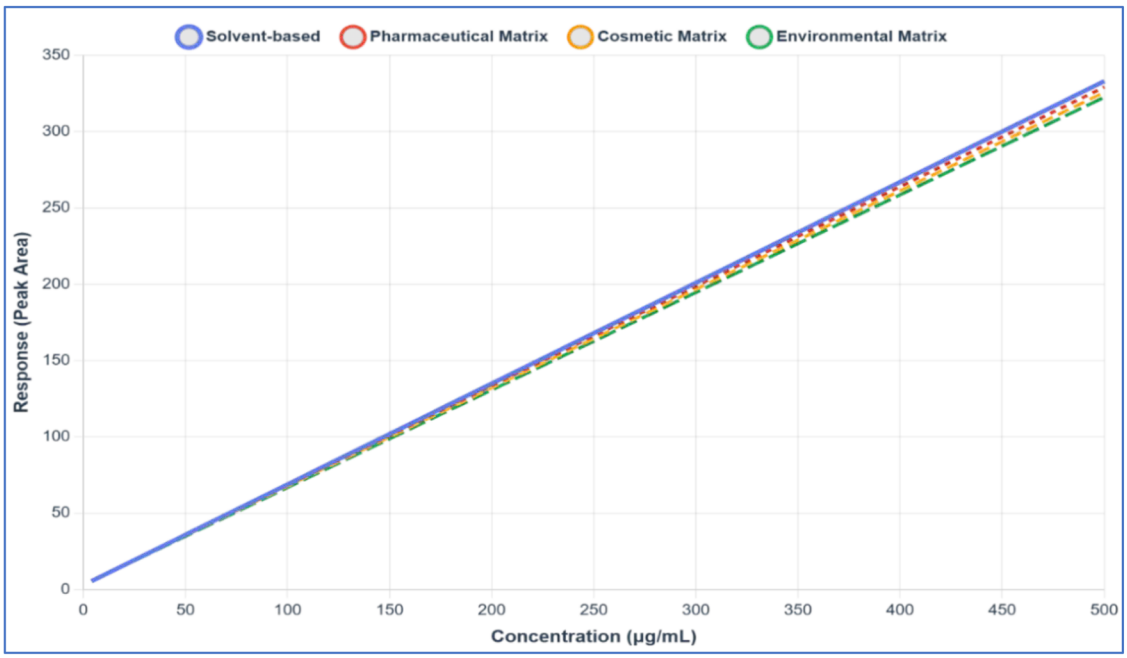
Accuracy

The accuracy of the developed method was evaluated through recovery studies conducted at three different concentrations. Known amounts of PVP K30 were spiked into pharmaceutical and cosmetic products as well as into environmental water samples. The method's performance was further evaluated by determining the percentage recovery. Representative chromatograms of the spiked samples are shown in Fig. 4 (A–F), while detailed recovery data are summarized in Tables 3 and 4.

Assays of Pharmaceutical and Cosmetic Products The method demonstrated excellent accuracy using matrix-matched calibration across multiple concentration levels in pharmaceutical and cosmetic matrices. Table 3 presents the recovery results obtained using matrix-matched quantification for two commercial products: Refresh® Classic eye drops and Got2B Glued Spiking



A. Calibration curve of PVP K30 showing the linear relationship between peak area and concentration (4–500 µg/mL).



B. Comparison of PVP K30 calibration curves showing matrix effects. All calibrations maintain excellent linearity ($R^2 > 0.999$) over 4-500 µg/mL with minimal matrix-induced signal suppression (1.24-3.24%) across pharmaceutical, cosmetic, and environmental samples.

Fig. 3 **A** Calibration curve of PVP K30 showing the linear relationship between peak area and concentration (4–500 µg/mL). **B** Comparison of PVP K30 calibration curves showing matrix effects. All calibrations maintain excellent linearity (R^2 >0.999) over 4–500 µg/mL with minimal matrix-induced signal suppression (1.24–3.24.24.24%) across pharmaceutical, cosmetic, and environmental samples

Table 2 Precision data for PVP K30 determination

Concentration (µg/mL)	Intraday Precision (RSD%, $n=6$)	Interday Precision (RSD%, $n=9$)
10 (low)	1.83	2.16
80 (medium)	1.42	1.87
400 (high)	0.95	1.23

Glue. Quantification was performed using matrix-matched calibration curves (Refresh® Classic: $y=0.6521x+3.1456$, $r^2 = 0.9995$; Got2B Glued: $y=0.6448x+3.2891$, $r^2 = 0.9993$). The chromatograms in Fig. 4 (A-B) illustrate spiked commercial samples at low concentration levels.

The nominal concentration for Refresh® Classic is based on the labeled amount (0.6% in 0.4 mL) diluted to 10 mL. The nominal concentration for Got2B Glued is based on the average amount found in the product after processing. The use of matrix-matched calibration resulted in excellent recovery percentages (99.4–99.7%) that meet ICH Q2(R1) acceptance criteria for pharmaceutical analysis. The RSD values remained excellent (<1.6%), confirming high precision and reproducibility across all spiking levels. These results demonstrate that the matrix-matched approach successfully compensates for minor matrix effects present in complex pharmaceutical and cosmetic formulations, ensuring accurate quantification for quality control applications.

Application To Environmental Water Samples The validated method was applied to environmental water samples using matrix-matched calibration curves prepared from blank environmental water processed through the complete SPE procedure. Table 4 summarizes the recovery results for

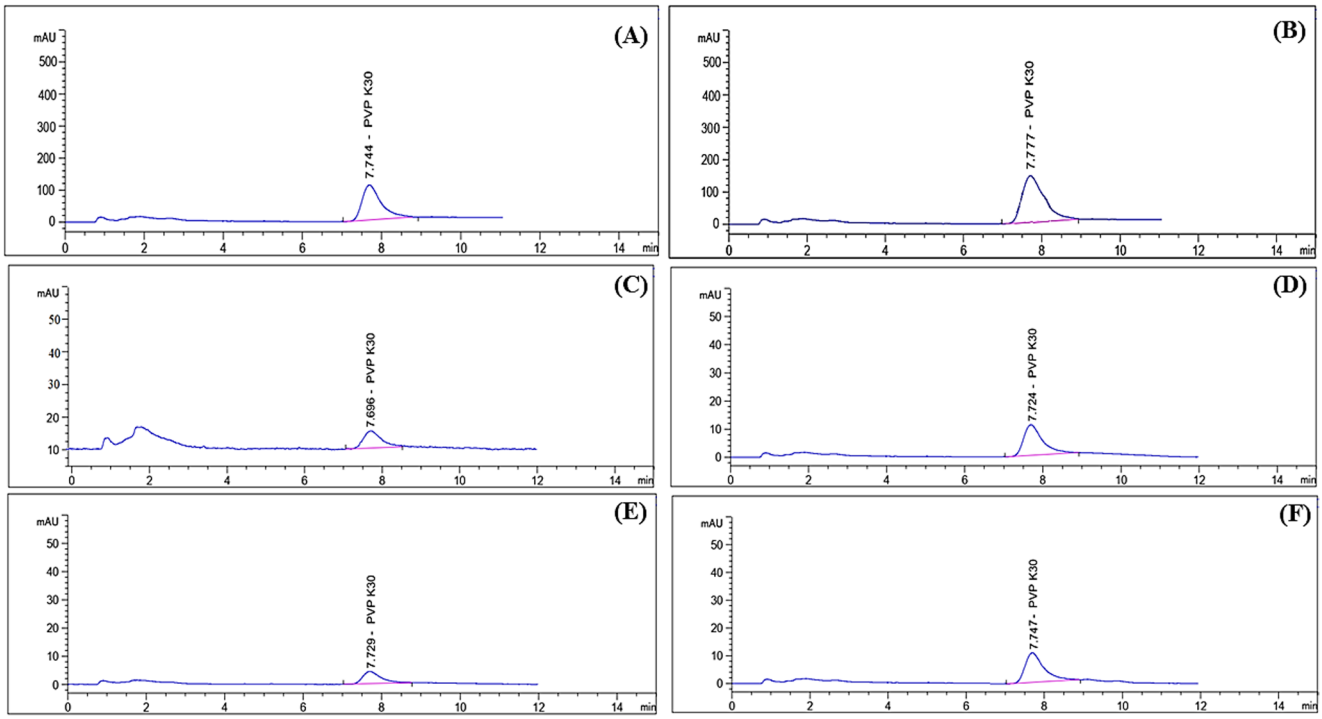


Fig. 4 HPLC chromatograms at low spiking levels for (**A**) Refresh® Classic eye drops, (**B**) Got2B-Glued Spiking Glue, (**C**) unspiked effluent (post-SPE), (**D**) spiked effluent (post-SPE), (**E**) unspiked influent (post-SPE), and (**F**) spiked influent (post-SPE)

Table 3 Accuracy and recovery data for the determination of PVP K30 in commercial products using matrix-matched calibration

Product	Nominal Conc. (µg/mL)	Spiking Level	Amount Added (µg/mL)	Amount Found (Mean±SD, µg/mL)	Recovery (%)	RSD (%, $n=3$)
Refresh® Classic	240	80%	192	190.8±2.9	99.4	1.52
		100%	240	238.9±3.2	99.5	1.34
		120%	288	287.1±3.8	99.7	1.32
Got2B Glued	230	80%	184	183.2±2.5	99.6	1.36
		100%	230	229.1±2.8	99.6	1.22
		120%	276	274.8±3.1	99.5	1.13

wastewater treatment plant (WWTP) influent and effluent samples collected from the Al-Gabal Al-Asfar plant. Quantification was performed using the matrix-matched calibration curve ($y = 0.6389x + 3.0124$, $r^2 = 0.9991$).

All reported influent and effluent concentrations represent the original environmental levels prior to sample preconcentration via solid-phase extraction (SPE). The SPE process achieved 500-fold enrichment (500 mL initial sample volume concentrated to 1 mL), enabling accurate quantification within the validated calibration range (4–500 µg/mL). The matrix-matched approach provided excellent recovery percentages for environmental samples (98.5–100.7.5.7%) that meet US EPA acceptance criteria for environmental analysis (80–120% recovery). This represents a significant improvement compared to solvent-based quantification and demonstrates the importance of accounting for matrix effects in complex environmental samples, even after SPE cleanup. The RSD values remained below 2.0%, indicating excellent precision and reproducibility despite the complexity of wastewater matrices. The chromatograms in Fig. 4 (C–F) illustrate both unspiked and spiked environmental samples after SPE processing, demonstrating the method's selectivity and sensitivity. These findings confirm the method's robustness, accuracy, and matrix tolerance, making it highly suitable for routine environmental monitoring and regulatory compliance applications related to polymer contaminants such as PVP K30.

Matrix Effects and Calibration Strategy Evaluation The systematic comparison between solvent-based and matrix-matched calibration approaches provided valuable insights into the method's performance across different matrix types. While the observed matrix effects were relatively minor (−1.24% to −3.24%), the matrix-matched approach demonstrated superior accuracy and reliability, particularly for environmental samples. Matrix effect assessment: The slight signal suppression observed across all matrices is consistent with the expected behavior of complex samples containing proteins, salts, surfactants, and other organic compounds that can interact with the analyte or compete for detector response. The pharmaceutical matrix showed the lowest matrix effect (−1.24%), likely due to the relatively simple aqueous formulation of eye drops with minimal interfering compounds. The cosmetic matrix exhibited moderate effects (−2.35%), reflecting the presence of polymeric excipients, styling agents, and preservatives that can cause minor signal suppression. The environmental matrix demonstrated the highest matrix effect (−3.24%), which is expected given the complexity of wastewater containing diverse organic and inorganic compounds, dissolved organic matter, and residual SPE matrix components. Calibration strategy justification: The matrix-matched calibration approach was selected over internal standard methodology for several scientifically sound reasons: Absence of suitable internal standard: No commercially available deuterated or isotopically labeled PVP K30 exists for this application. Alternative polymeric internal standards (such as other grades of PVP or structurally similar polymers) would require separate method development and validation, and may not adequately compensate for PVP K30-specific matrix interactions due to differences in molecular weight distribution and chemical behavior. Matrix diversity challenge: The three target matrices (pharmaceutical, cosmetic, environmental) have vastly different chemical compositions and potential interferents, making it unlikely that a single internal standard would provide appropriate and consistent compensation across all applications. Each matrix type presents unique challenges that are better addressed through matrix-specific calibration. Practical implementation advantages: Matrix-matched calibration is more straightforward to implement in routine analytical laboratories, requires no additional specialized standards, and provides matrix-specific compensation without introducing method complexity or additional validation requirements. Regulatory alignment: Matrix-matched calibration is widely accepted and recommended by regulatory agencies (ICH, FDA, EPA) for complex matrix analysis, particularly when suitable internal standards are not available. Performance improvement with matrix-matched calibration: The quantitative benefits of the matrix-matched approach are clearly demonstrated by the improved recovery rates: environmental samples showed improvement from 82.0 to 85.1% (solvent-based) to 98.5–100.7.5.7% (matrix-matched), while pharmaceutical and cosmetic products achieved near-ideal recoveries (99.4–99.7%). This improvement is particularly

Table 4 Accuracy and recovery data for the determination of PVP K30 in environmental water samples using matrix-matched calibration

Sample Type	PVP K30 Found (µg/L)	Spiking Level	Spiked Amount (µg/L)	Measured Concentration (µg/L)	Recovery (%)	RSD (%, $n = 3$)
WWTP Influent	42.75 ± 0.80	Low	20	62.89	100.7	1.85
		Medium	40	82.15	98.5	1.92
		High	80	121.95	99.0	1.78
WWTP Effluent	18.34 ± 0.35	Low	20	38.12	99.0	1.95
		Medium	40	57.89	98.9	1.88
		High	80	97.78	99.3	1.82

critical for environmental monitoring applications where accurate quantification at trace levels is essential for regulatory compliance, risk assessment, and environmental protection. The matrix-matched calibration strategy ensures that any suppression or enhancement effects present in real samples are equally present in the calibration standards, providing automatic compensation and eliminating systematic bias. This approach validates the method's reliability across the diverse range of matrices encountered in PVP K30's product-to-environment lifecycle and confirms its suitability for both industrial quality control and environmental monitoring applications.

Selectivity

The developed method demonstrated excellent specificity for PVP K30. No interfering peaks were observed at the retention time of PVP K30 in any of the tested blanks. Analysis of blank samples, including distilled water, the mobile phase, and unspiked WWTP influent and effluent samples (containing PVP K30 below the LOD), revealed no significant coelution with the analyte of interest. This confirms the method's selectivity and robustness in complex environmental matrices, as illustrated in Fig. 5A.

Robustness

The method's robustness was assessed by deliberately varying critical parameters and evaluating their impact on the analytical performance by statistically comparing the results to those obtained under the normal conditions. A standard solution of PVP K30 (200 µg/mL) was analyzed under the normal conditions (flow rate: 1.5 mL/min; detection wavelength: 210 nm; mobile phase ratio: 50:50 v/v) and under each modified condition. The retention time and peak area were recorded for six replicates per condition.

The results were statistically evaluated using a two-tailed t-test (for comparing means) and an F-test (for comparing variances) at a 95% confidence level ($\alpha = 0.05$). The calculated p-values for the t-test and F-test are presented in Table 5. A p-value greater than 0.05 indicates that there is no statistically significant difference between the results obtained under normal and modified conditions, confirming the method's robustness to that parameter variation. Crucially, the relative standard deviation (RSD%) of the peak area for each set of six replicates was calculated to validate the precision of the measurement under each modified condition.

The method demonstrated acceptable robustness, with key parameters showing predictable behavior under deliberate variations. Changes in flow rate (±0.1 mL/min) and

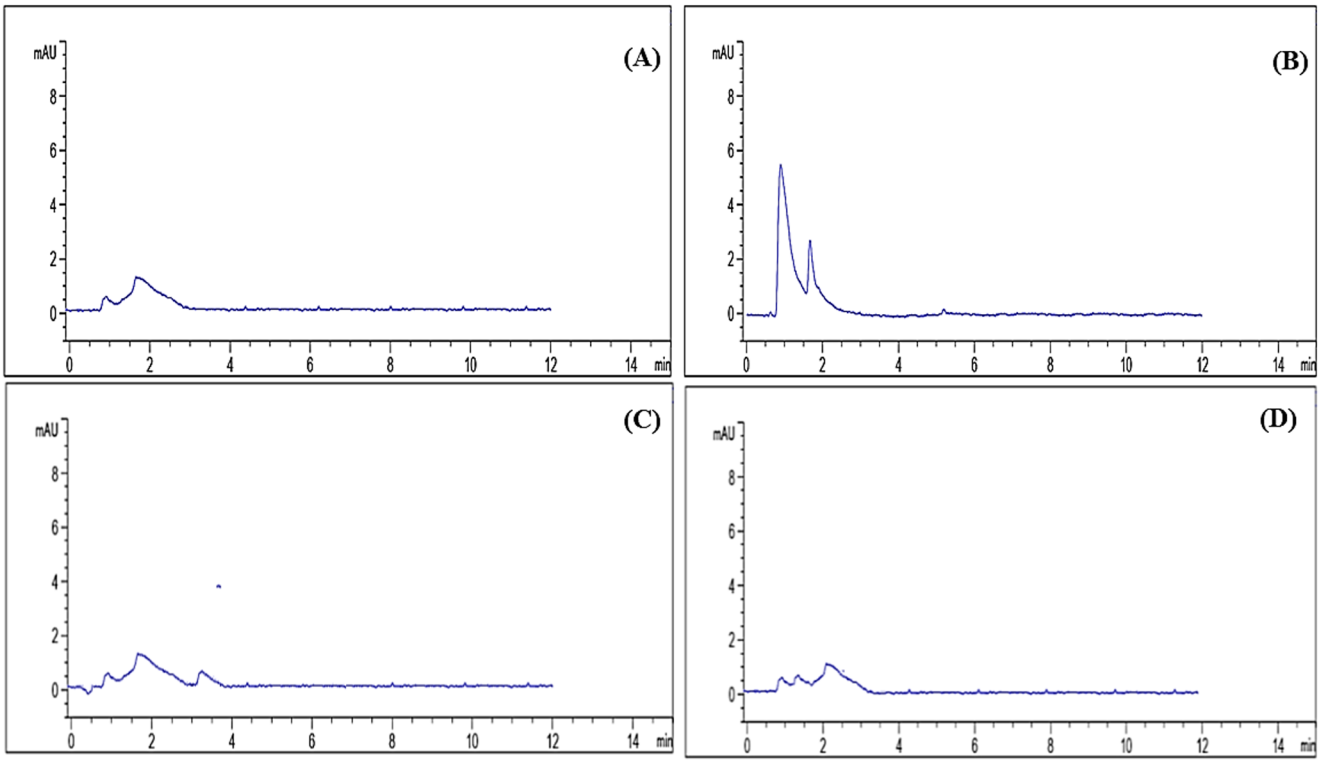


Fig. 5 Representative chromatograms demonstrating the specificity of the developed method for PVP K30: (**A**) distilled water, (**B**) mobile phase, (**C**) unspiked WWTP influent sample (pre-SPE), and (**D**) unspiked WWTP effluent sample (pre-SPE). No interfering peaks were observed at the retention time of PVP K30, confirming method selectivity

Table 5 Statistical evaluation of robustness for PVP K30 determination, including peak area RSD%

Parameter	Variation	Retention Time (min) Mean ± SD	Peak Area (AU) Mean ± SD	Peak Area RSD%	t-test *p*-value (RT)	F-test *p*-value (RT)	t-test *p*-value (Area)	F-test *p*-value (Area)	Conclusion
Normal Condition	-	7.70 ± 0.05	134.528 ± 1.205	0.90%	-	-	-	-	-
Flow Rate	+0.1 mL/min (1.6 mL/min)	6.84 ± 0.06	133.891 ± 1.567	1.17%	< 0.001	0.321	0.421	0.198	Significant change in RT (expected), no significant change in Area or its variance ($p > 0.05$). Excellent precision (RSD% = 1.17%).
	−0.1 mL/min (1.4 mL/min)	8.35 ± 0.07	135.102 ± 1.432	1.06%	< 0.001	0.105	0.385	0.275	Significant change in RT (expected), no significant change in Area or its variance ($p > 0.05$). Excellent precision (RSD% = 1.06%).
Detection Wavelength	+2 nm (212 nm)	7.69 ± 0.06	127.655 ± 1.805	1.41%	0.725	0.421	**< 0.05**	0.087	No significant change in RT or its variance. Significant change in mean Area ($p < 0.05$) due to reduced sensitivity at this wavelength, though precision remains excellent (RSD% = 1.41%).
	−2 nm (208 nm)	7.71 ± 0.05	129.342 ± 1.621	1.25%	0.681	1.000	**< 0.05**	0.352	No significant change in RT or its variance. Significant change in mean Area ($p < 0.05$) due to reduced sensitivity at this wavelength, though precision remains excellent (RSD% = 1.25%).
Mobile Phase Ratio	+1% B (51% ACN)	7.02 ± 0.08	135.855 ± 1.723	1.27%	< 0.001	0.058	0.061	0.175	Significant change in RT (expected), no significant change in Area or its variance ($p > 0.05$). Excellent precision (RSD% = 1.27%).
	−1% B (49% ACN)	8.11 ± 0.09	133.205 ± 1.654	1.24%	< 0.001	0.052	0.055	0.221	Significant change in RT (expected), no significant change in Area or its variance ($p > 0.05$). Excellent precision (RSD% = 1.24%).

SD: Standard Deviation; ACN: Acetonitrile; RT: Retention Time; AU: Arbitrary Units

mobile phase composition (± 1% acetonitrile) caused statistically significant but expected retention time shifts ($p < 0.001$) while maintaining excellent peak area stability ($p > 0.05$) with all RSD% values below 1.3%. Although a ± 2 nm wavelength variation produced a statistically significant sensitivity change in peak area ($p < 0.05$), measurement precision remained excellent (RSD% <1.5%) and not significantly different from normal conditions (F-test $p > 0.05$). All RSD% values remained well below the 2.0% acceptance criterion, confirming the method's reliability and precision despite minor operational fluctuations.

Ruggedness

The results of the ruggedness study are summarized in Table 6. The RSD values for the retention time, peak area, and theoretical plates between the two analysts and instruments were all below 2.0%. This low variability demonstrates that the method is insensitive to minor but expected changes in operational conditions and is highly reproducible. The successful execution of this study confirms the method's ruggedness and its suitability for routine application in different laboratory environments.

Table 6 Results of the ruggedness study

Parameter	Analyst 1 ($n = 6$) Agilent 1200	Analyst 2 ($n = 6$) Agilent 1260	RSD (%) Between Sets
Retention Time (min)	7.71 ± 0.04	7.68 ± 0.05	0.45%
Peak Area	134.528 ± 1.205	132.951 ± 1.878	1.18%
Theoretical Plates	2550 ± 95	2489 ± 112	1.72%

Data presented as Mean ± Standard Deviation

Method Comparison with Existing Techniques

The developed mixed-mode HPLC method was statistically compared with previously reported techniques for PVP analysis to quantitatively demonstrate its improved performance. A common challenge in polymer analysis is the direct comparison of methods designed for different purposes. While other methods may be faster and avoid pretreatment for quantifying PVP in concentrated pharmaceutical products [20]; [27]; [54], they lack the sensitivity for environmental monitoring. The key innovation of our work is the development of a single, validated method that

spans this entire application range, from quality control of products to trace-level environmental analysis, which is unprecedented. The comparison focused on the limits of detection (LOD) and quantification (LOQ) as key metrics of analytical sensitivity. The Fold Improvement for each parameter was calculated as follows:

$$FoldImprovement = (ExistingMethodValue)/(ProposedMethodValue).$$

Table 7 summarizes the comparative performance parameters and the calculated fold improvements. Our method demonstrates a 4.2-fold improvement in LOD and a 3.9-fold improvement in LOQ compared to the most sensitive previously reported method [27]. Furthermore, and most critically, it is the only method that achieves the sensitivity required for environmental monitoring (LOD: 2.4 ng/L) through its optimized SPE protocol. This preconcentration step is not a limitation but the fundamental enabling factor that permits the quantification of PVP at trace levels in water, a feat that the other compared methods cannot accomplish. The other methods are incapable of detecting PVP at the ng/L to µg/L levels typically found in environmental samples [7–12]. The 12-minute analysis time is competitive, and the isocratic elution simplifies method transfer and reduces instrumental complexity. This comprehensive comparison underscores that the true advantage of our method is not just its sensitivity in a pure standard, but its unparalleled versatility and effectiveness across the three distinct and challenging matrices (pharmaceutical, cosmetic, environmental) that define the lifecycle of PVP, fulfilling a significant gap in the current analytical landscape

LOD/LOQ values for the present method's environmental application are given in ng/L in parentheses, representing the sensitivity in the original water sample after SPE preconcentration.

Greenness and Practicality Assessment of the Proposed Method

The sustainability and practical applicability of the newly developed analytical methodology were thoroughly evaluated using multiple modern assessment tools. This evaluation was conducted to provide a transparent, quantitative audit of the method's environmental footprint and practicality, acknowledging the necessary compromises made to achieve its core analytical objectives of sensitivity, versatility, and robustness for a challenging polymer analyte. Several assessment tools, including AGREE, AGREEprep, MoGAPI, BAGI, and RGB12, were employed to systematically examine the ecological impact and practical utility of our analytical approach [56–58].

The first assessment tool utilized is AGREE (Analytical Greenness metric), which evaluates compliance with the 12 principles of green analytical chemistry. The overall score of our method is visually represented at the center of a circular pictogram (Fig. 6), providing a quantitative measure of its environmental performance [56, 59].

AGREEprep was used to specifically assess the sample preparation procedure against the 10 principles of green sample preparation. The overall AGREEprep score for the method was calculated to be 0.52 (Fig. 7), reflecting the environmental impact of the sample preparation process [58].

The Modified Green Analytical Procedure Index (MoGAPI) provided a detailed evaluation of the entire analytical procedure (Fig. 8). This tool combines visual assessment with quantitative scoring, offering comprehensive insight into the method's environmental impact across all stages [56, 60].

The Blue Applicability Grade Index (BAGI) was applied to assess the practical aspects of the method. As depicted in Fig. 9, our method achieved a BAGI score of

Table 7 Statistical comparison with existing PVP analytical methods

Method	Column Type	Detection	Sample Prep	Analysis Time (min)	Linear Range	LOD (µg/mL)	LOQ (µg/mL)	Fold Improvement (LOD)	Fold Improvement (LOQ)	Primary Application	Reference
Present method	Hi-Plex H	UV 210 nm	Dilution or SPE	~ 12	4–500 µg/mL	1.2	3.8	-	-	Multimatrix (Pharma, Cosmetic, Environmental)	This work
(After SPE)						(2.4 ng/L)	(7.6 ng/L)				
SEC-UV	TSK-Gel	UV 220 nm	Dilution	~ 15	5–100 µg/mL	5.0	15.0	4.2	3.9	Ophthalmic Solutions	[22]
RP-HPLC	C18	UV 205 nm	Dilution	~ 10	8.5–100 µg/mL	2.8	8.5	2.3	2.2	Pharmaceutical	[29]
RP-HPLC	C18	UV 200 nm	Dissolution	~ 6	50–500 µg/mL	16.0	50.0	0.075	0.076	Adhesives	[51]
SEC-HPLC	Ultrahydrogel	RI	Dilution	~ 20	Not specified	35.5	107.5	0.034	0.035	Pharmaceutical	[53]

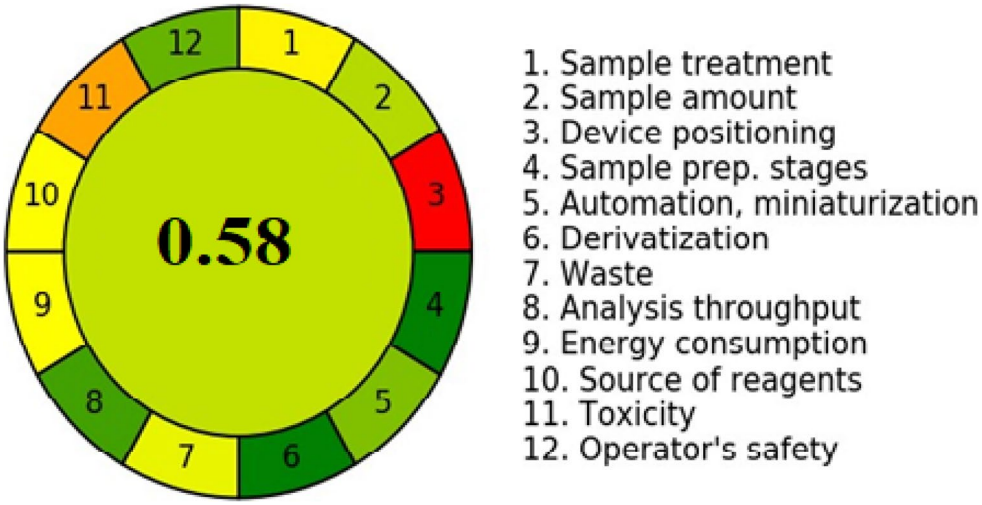


Fig. 6 The AGREE method



Fig. 7 The AGREEprep method

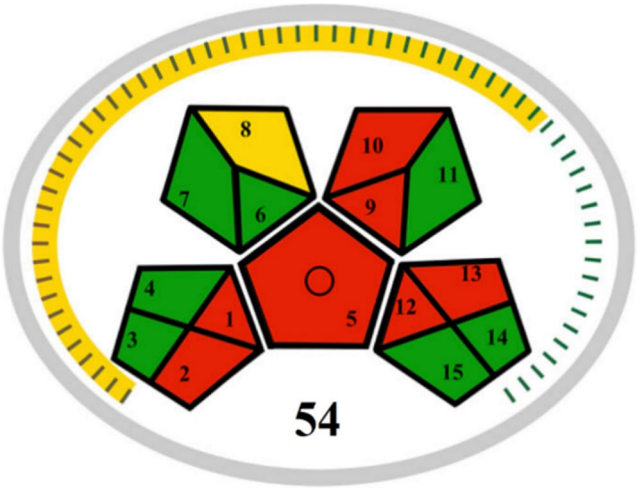


Fig. 8 The MoGAPI method

70.0, highlighting its excellent practical applicability, good throughput potential, and operational efficiency [32].

Finally, the RGB12 algorithm was used to calculate an overall whiteness score (Fig. 10) of 72.0, integrating analytical efficacy (red), ecological impact (green), and practical/economic feasibility (blue) into a single sustainability metric [33, 56].

Comparative Greenness Assessment

To objectively benchmark the environmental performance of our method against existing literature, a comparative analysis was conducted using the AGREE metric [61]. We calculated the AGREE scores for our proposed mixed-mode HPLC method and two representative HPLC methods from the literature previously used for PVP analysis [27].

The results, summarized in Table 8, demonstrate a clear and significant advancement in greenness. The method by Jones et al. (2004) [27], which utilizes an RP-C18 column with a mobile phase of propanol-water containing 0.01% TFA, achieved an AGREE score of 0.50. The method by Gupta et al. (2024) [54], also using a C18 column but with a methanol-water mobile phase, scored 0.48. In contrast, our method achieved a substantially higher score of 0.58. This improvement can be attributed to several features informed by green analytical chemistry principles, including the use of a mixed-mode column that operates effectively with simpler isocratic elution, reduced total solvent consumption compared to gradient methods, and careful optimization leading to shorter analysis time [61, 62].



Fig. 9 BAGI method

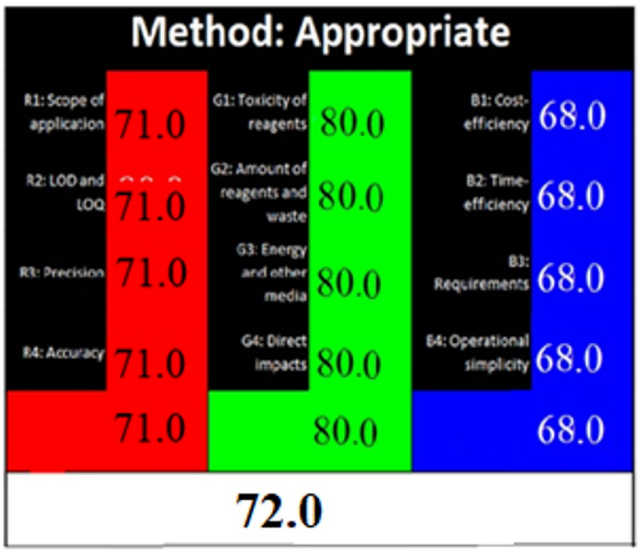


Fig. 10 The RGB-12 method

This holistic evaluation confirms that the method achieves a functional balance between essential analytical performance and sustainability principles. The AGREE score of 0.58 provides a quantitative baseline that reflects the environmental cost of using necessary but hazardous reagents to solve a challenging analytical problem. This score, combined with the strong BAGI rating, portrays a method that is highly practical and represents a more sustainable option within its specific analytical niche, while openly acknowledging its limitations [62–65].

Table 8 Comparative AGREE greenness assessment with previously published HPLC methods for PVP analysis

Method Description	Mobile Phase	AGREE Score	Key Greenness Limitations	Reference
Proposed Mixed-Mode HPLC	Dilute Perchloric Acid: ACN (1:1)	**0.58**	Use of hazardous perchloric acid and acetonitrile	**This study**
RP-HPLC for PVP K15	Propanol: Water, 0.01% TFA	0.50	Use of less green propanol and TFA modifier	[27]
RP-HPLC for PVP in Adhesives	Methanol: Water	0.48	Higher organic solvent consumption, no waste consideration	[54]

Study Limitations and Future Perspectives

Despite the demonstrated sensitivity, accuracy, and applicability of the developed method across pharmaceutical, cosmetic, and environmental samples, several limitations should be acknowledged. The current study focused exclusively on PVP K30, and the method's performance for other polymeric compounds or different grades of polyvinylpyrrolidone has not yet been established. Additionally, environmental sampling was limited to wastewater influent and effluent from selected locations, which may not fully capture the variability of polymer contamination in diverse environmental settings.

Although the solid-phase extraction (SPE) procedure has proven effective, further optimization may be needed to improve recovery efficiency, particularly for trace-level detection in complex environmental matrices containing various organic and inorganic components. A key priority for future work will be the re-investigation and optimization of the SPE elution protocol to replace methanol with a greener solvent, such as ethanol, without compromising the excellent recovery rates achieved in this study. Moreover, long-term method stability and interlaboratory reproducibility, which are essential for confirming its robustness for routine environmental monitoring, have not yet been assessed.

Future work should focus on extending the method to detect a broader range of polymeric contaminants and exploring coupling with advanced analytical techniques, such as high-resolution mass spectrometry, to increase detection limits and enable structural analysis. Additionally, applying this method in conjunction with ecotoxicological evaluations could provide deeper insight into the environmental behavior and potential impacts of PVP K30 and related polymers.

Overall, continued refinement and validation of this method, alongside expanded applications, will increase its utility for environmental risk assessment and support improved management of polymeric pollutants in aquatic ecosystems.

Conclusion

This study successfully developed and validated a novel, sensitive, and reliable mixed-mode HPLC method utilizing a Hi-Plex H column for the quantification of polyvinylpyrrolidone K30 (PVP K30) in pharmaceutical, cosmetic, and environmental water samples. The method demonstrated excellent accuracy, precision, and robustness across a wide concentration range. A key finding was that the mixed-mode mechanism provided a symmetric peak shape (Tf = 1.12) for the polydisperse polymer, which is critical for reliable integration and quantification.

The incorporation of an optimized solid-phase extraction protocol enables effective 500-fold preconcentration of environmental samples, which is the critical factor that allows for reliable quantification at trace levels (ng/L) in complex wastewater matrices. This addresses a fundamental weakness of existing methods, which are limited to high-concentration product analysis.

The method's successful validation and application underscore its potential as a valuable tool for quality control in the pharmaceutical and cosmetic industries, as well as for environmental monitoring and risk assessment of polymer contaminants. Its unique ability to seamlessly span from percent-level formulations to part-per-trillion environmental residues within a single analytical framework represents a significant advancement.

Finally, the sustainability and practicality profile of the methodology was transparently assessed using the AGREE, AGREEprep, MoGAPI, BAGI, and RGB12 tools. This multi-faceted assessment provides a benchmark for the environmental impact of analyzing PVP K30 and clearly identifies aspects, such as solvent use, where future work should focus to develop greener alternatives. The method stands as a sensitive, versatile, and practical solution for monitoring PVP K30 across its product-to-environment lifecycle.

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Author Contributions Author ContributionsSamar M. Mahgoub: Conceptualization, Methodology, Supervision, Writing – Review & Editing.Hassan A. Rudayni and ahmed allam: Method Validation, Data Curation, Formal Analysis.Abdelatty M. Radalla and Heba Ahmed: Investigation, Resources, Visualization.Rehab Mahmoud: Experimental Work, Writing – Original Draft, Project Administration, Corresponding Author.All authors have read and approved the final manuscript.

Data Availability The data was used for the research described in the article are available from the corresponding author upon the request for scientific issues.

Declarations

Consent for Publication The authors confirm:

That the work described has not been published before.That it is not under consideration for publication elsewhere.That its publication has been approved by all coauthors.

Ethics Approval and Consent to Participate Not applicable, as the study did not apply to human or animal studies. The article does not include any studies on human participants or animals conducted by any of the authors.

Competing Interests The authors declare no competing interests.

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